

# Legal Metrology — Centralized Database Architecture

Single Source of Truth PostgreSQL 16 + PostGIS Entity-Relationship Specification (45 Tables)

|                                          |                                       |                             |                                    |
|------------------------------------------|---------------------------------------|-----------------------------|------------------------------------|
| Database Engine: PostgreSQL 16 + PostGIS | Object Storage: MinIO / S3            | Graph DB: Neo4j 5.x         | Total Tables: 45 Relational Tables |
| Backend API: NestJS (TypeScript)         | AI/OCR Engine: FastAPI (Vision + NER) | Auth: Keycloak (JWT + RBAC) | Status: 100% Live & Seeded         |

## System Entity Domains & Relational Schema Dictionary

| Domain Area              | Core Entities & Tables                                                              | Primary Keys & Foreign Keys                                                                                 | Key Constraints & Relational Purpose                                                                    |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1. User & Access Control | roles<br>users<br>inspectors<br>controllers<br>consumers<br>audit_logs              | role_id (PK)<br>user_id (PK, role_id FK)<br>inspector_id (PK, user_id FK)<br>controller_id (PK, user_id FK) | Manages Keycloak JWT auth sync and RBAC permissions. Record-level audit tracking in audit_logs.         |
| 2. Business Master       | businesses<br>business_identifiers<br>business_locations                            | business_id (PK)<br>identifier_id (PK, business_id FK)<br>location_id (PK, business_id FK)                  | Strictly identifies corporate entities via UNIQUE GSTIN and PAN. PostGIS coordinates for geo-tracking.  |
| 3. Product Master        | products<br>product_identifiers<br>product_categories<br>product_business_rel       | product_id (PK, category_id FK)<br>identifier_id (PK, product_id FK)<br>category_id (PK)                    | Tracks commodity identity using UNIQUE GTIN, Barcode, and QR codes. Enables statewide product history.  |
| 4. Rules & Compliance    | laws<br>rules<br>rule_conditions<br>compliance_requirements<br>identifier_val_rules | law_id (PK)<br>rule_id (PK, law_id FK)<br>condition_id (PK, rule_id FK)                                     | Data-driven legal rules (PCR 2011 & LMA 2009). Evaluates severity: MINOR, MODERATE, CRITICAL, SEVERE.   |
| 5. Inspection & Evidence | inspections<br>inspection_products<br>evidence<br>ocr_results<br>extracted_fields   | inspection_id (PK)<br>evidence_id (PK, inspection_id FK)<br>ocr_id (PK, evidence_id FK)                     | Inspector field work orders. MinIO S3 object keys with SHA-256 hashes. Raw OCR text & canonical fields. |

|                                |                                                                                                               |                                                                                                  |                                                                                                                |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 6. Consumer Complaints         | complaints<br>complaint_evidence<br>complaint_status_history                                                  | complaint_id (PK, consumer_id FK)<br>complaint_evidence_id (PK)                                  | Public citizen reporting module. Generates UNIQUE complaint tracking IDs and triggers inspection leads.        |
| 7. AI Predictions & Violations | ai_predictions<br>ai_feedback<br>violations                                                                   | violation_id (PK, inspection_id FK, rule_id FK,<br>evidence_id FK)                               | Human-in-the-Loop AI: Predictions generate status='POTENTIAL'. Inspector explicitly confirms/rejects.          |
| 8. Enforcement & Notices       | notices<br>notice_documents<br>seizure_bills<br>panchanamas<br>improvement_notices<br>compounding_orders      | notice_id (PK, violation_id FK)<br>seizure_id (PK)<br>improvement_id (PK)<br>compounding_id (PK) | Digital generation of Show Cause Notices, Seizure Bills, Improvement Notices (for MINOR errors) & Compounding. |
| 9. Supply Chain Intelligence   | suppliers<br>purchase_bills<br>purchase_bill_items<br>supply_chain_relationships<br>supply_chain_transactions | bill_id (PK)<br>relationship_id (PK, upstream FK, downstream FK)                                 | Invoice parsing & multi-hop supply chain trace (Retailer -> Distributor -> Manufacturer). Synced to Neo4j.     |

## Key ER Diagram Relationships Summary

- **ROLES (1) ■■■ (N) USERS:** User profile access level.
- **BUSINESSES (1) ■■■ (N) BUSINESS\_IDENTIFIERS:** 1-to-N lookup for GSTIN and PAN.
- **PRODUCTS (1) ■■■ (N) PRODUCT\_IDENTIFIERS:** 1-to-N lookup for GTIN, Barcode, and QR code.
- **INSPECTIONS (1) ■■■ (N) EVIDENCE ■■■ (1) ■■■ (N) OCR\_RESULTS ■■■ (N) EXTRACTED\_FIELDS:** OCR extraction pipeline.
- **VIOLATIONS (1) ■■■ (N) NOTICES ■■■ (1) ■■■ (1) SEIZURE\_BILLS / IMPROVEMENT\_NOTICES / COMPOUNDING\_ORDERS:** Enforcement workflow.
- **BUSINESSES (N) ■■■ (N) SUPPLY\_CHAIN\_RELATIONSHIPS:** Multi-hop upstream graph traceability.